Systematic Literature Review Protocol

Optimisation of Large Language Models on Resource-Constrained Devices (2024–2026)

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1. Background and objective

Deploying large language models (LLMs) on edge devices is limited by memory, compute, latency and energy constraints. This review identifies and synthesises the optimisation techniques proposed between 2024 and 2026 to run LLMs on resource-constrained hardware, the platforms used to validate them, their measured effect on physical resources, and the remaining research gaps.

The review follows the guidelines of Kitchenham and Charters (2007). PICOC is used to decompose the objective into search terms, and the selection flow is reported according to PRISMA 2020, which acts as a reporting guideline and does not replace the Kitchenham–Charters procedure:

> Kitchenham–Charters → PICOC → search and eligibility → PRISMA → quality assessment → extraction and synthesis

2. Research questions

ID	Question
RQ1	Which optimisation techniques have been used between 2024 and 2026 to enable efficient deployment of LLMs on resource-constrained devices?
RQ2	Which edge hardware platforms and architectures have been used to empirically validate these techniques?
RQ3	How do the techniques affect the main resource constraints, namely memory, latency, energy consumption and model performance?
RQ4	Which relationships can be identified between optimisation techniques, hardware platforms and efficiency metrics, and which trade-offs between resource reduction and model performance are observed?
RQ5	Which methodological limitations and research gaps remain for the validation and efficient deployment of LLMs on edge devices?

3. PICOC

Element	Definition	In string
Population	Large language models and foundation models running on constrained hardware	Yes (block 1)
Intervention	Algorithmic optimisation: quantisation, pruning, knowledge distillation, model compression, low-bit inference	Yes (block 2)
Comparison	Unoptimised base models or cloud deployment	No (to maximise recall)
Outcomes	Computational efficiency, latency, memory, energy, accuracy retention	No (to maximise recall)
Context	Edge computing, mobile and resource-constrained devices, FPGAs, smartphones	Yes (block 3)

4. Search strategy

- **Sources:** Scopus and Web of Science Core Collection. Both are selective multidisciplinary indexes that aggregate IEEE Xplore, ACM Digital Library, Elsevier, Springer and MDPI content. Access was limited to these two databases (declared limitation).
- **Query** (identical in both databases):

```
("Large language model" OR "LLM*" OR "Foundation model*")
AND ("quantization" OR "model compression" OR "low-bit"
     OR "knowledge distillation" OR "pruning")
AND ("edge computing" OR "mobile" OR "resource-constrained"
     OR "edge device*" OR "FPGA" OR "smartphone")
```

- **Limits:** no date or document-type restriction at search time; both are applied afterwards through the eligibility criteria. The exact string and record counts are kept in `docs/search_strings.txt`.
- **Records retrieved:** 724 (Scopus 586, Web of Science 138). Data collection covers literature published up to mid-2026.

5. Eligibility criteria

Inclusion (IC)	Exclusion (EC)
IC-1 Studies centrally focused on the optimisation of LLMs.	**EC-1** Studies exclusively targeting unconstrained cloud environments (cloud used only as a baseline is acceptable).
IC-2 Empirical validation oriented to resource-constrained environments, either on identifiable edge hardware or through compression experiments aimed at the edge.	**EC-2** Purely theoretical proposals without experimental validation on language models.
IC-3 Primary research articles in peer-reviewed journals reporting definitive results and quantitative metrics.	**EC-3** Documents without verifiable results or without quantitative efficiency or performance metrics.
IC-4 Written in English or Spanish.	**EC-4** Grey or secondary literature: reviews, surveys, conference proceedings, book chapters, theses, editorials.

IC-2 is checked against what each primary study declares, so studies with edge-oriented validation are retained even when they do not name a concrete device. This distinction is kept explicit in the synthesis (Group A vs. Group B, Section 8).

6. Selection procedure

Criteria are applied sequentially. Every count below is recomputed from the raw exports by `scripts/prisma_flow.py`.

Stage	Removed	Remaining
Records identified (Scopus 586 + WoS 138)	–	724
Duplicates (normalised title; WoS metadata kept)	171	553
Non-primary document type (EC-4)	385	168
Language not covered (IC-4)	5	163
Untyped secondary literature or incomplete metadata	14	149
Excluded on title and abstract (mainly IC-2 / EC-1)	69	80
Excluded at full text (do not optimise or evaluate an LLM)	7	**73**

Decisions and reasons for each record are stored in `data/screening_title_abstract.csv` and `data/fulltext_exclusions.csv`.

7. Quality assessment

Each included study is scored on three questions with an ordinal scale (1.0 = fully met, 0.5 = partially met, 0.0 = not met):

- **QA1 (hardware)** Does the study explicitly specify the hardware architecture and physical deployment environment? 1.0 = a concrete device, board or SoC is named; 0.5 = only a generic descriptor such as “edge devices”, “smartphones” or “GPU”; 0.0 = hardware not declared.
- **QA2 (method)** Is the proposed optimisation method reproducible?
- **QA3 (metrics)** Are quantitative comparative metrics reported against established reference models?

The assessment is **descriptive only**: no study was excluded because of its score and the synthesis is not weighted by it. Scores are in `data/quality_assessment.csv` (Annex B).

8. Data extraction

For every included study the following fields were extracted (`data/corpus_73.csv`, Annex A):

- Bibliographic metadata: authors, title, year, journal, DOI, citations, keywords, affiliations, source database.
- **Optimisation technique** as declared by the study, and its coding into four non-exclusive families: quantisation, pruning/sparsity, low-rank adaptation (LoRA/PEFT/factorisation) and knowledge distillation.
- **Hardware** exactly as declared, without completing it with assumptions, and its coding into one platform class: NVIDIA Jetson, ARM SoC / Raspberry Pi / NPU, server FPGA, desktop/server GPU, or no edge platform.
- **Main result**: latency, memory and accuracy-retention figures reported.

Group A = studies that identify a concrete edge platform (Jetson or ARM SoC class) *and* report physical metrics measured on it. **Group B** = studies that report metrics without specifying an edge platform. Data-centre FPGAs and desktop/server GPUs are kept outside Group A because their power budgets exceed edge constraints. Hardware was first coded from title, abstract and keywords and then checked against the full text of studies reporting metrics.

9. Synthesis

Because models, platforms, experimental set-ups and metrics are highly heterogeneous, no statistical meta-analysis is performed. Synthesis is descriptive and thematic, organised by research question:

1. RQ1: frequency of optimisation families.
2. RQ2: distribution of validation platforms.
3. RQ3: reported effects on memory, latency, energy and accuracy.
4. RQ4: technique × platform matrix and observed trade-offs.
5. RQ5: limitations and gaps derived from the above.

A descriptive bibliometric characterisation of the corpus (years, journals, Bradford zones, Lotka distribution, citations) is reported for context only and is not used to infer quality or impact.

10. Deviations and limitations

- Only Scopus and Web of Science were available to the authors.
- EC-4 removed conference proceedings, which are a primary publication channel in machine-learning systems.
- LoRA/PEFT terms were not part of the search string; low-rank studies were retrieved only when indexed under compression or deployment terms.
- Seven studies passing title/abstract screening were excluded at full text because they do not optimise or evaluate an LLM (ResMLP image classification, IVF-PQ index search and construction on FPGA, generic DNN swapping, event-camera depth estimation, an industrial multimodal copilot and speech-translation models on an NPU).